Unit 2: 1D. Module 1

D.1. Measurement Concepts: Part 1

D.3. Overhead Costs: Part 1

Cost Objects (or Objectives)

D.1. Measurement Concepts: Part 1
D.3. Overhead Costs: Part 1

Cost Objects

~~Cost objects (objectives) are resources or activities that serve as the basis for management decisions.~~

- Product costs (manufacturing costs)
  - Direct materials (DM), direct labor (DL), manufacturing overhead (OH)
  - Prime costs = DM + DL; Conversion costs = DL + OH
  - Treated as inventory and part of cost of goods sold when sold
- Period costs (nonmanufacturing costs)
  - Selling, general, and administrative (SG&A) expenses and interest expenses
  - Treated as expenses in the period incurred

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Cost Pools and Cost Assignments

D.1. Measurement Concepts: Part 1
D.3. Overhead Costs: Part 1

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Cost Pools and Cost Assignments

- Cost pools are collections of cost elements assigned to cost objects using a common basis of allocation.
- Costs are assigned to cost pools using direct tracing (direct costs) or allocation (indirect costs).
- Direct costs are easily traceable to a cost pool or object.
  - Direct materials (cost of materials purchased and used in production)
  - Direct labor (cost of labor related to production)
- Indirect (overhead) costs are not easily traceable to a cost pool or object and are typically incurred to benefit two or more cost pools or objects.
  - Indirect materials (e.g., cleaning supplies, safety equipment)
  - Indirect labor (e.g., QA staff, maintenance workers)
  - Other indirect costs (e.g., rent, property taxes, insurance, utilities)

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Cost Behavior (Fixed vs. Variable)

D.1. Measurement Concepts: Part 1
D.3. Overhead Costs: Part 1

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Cost Behavior (Fixed vs. Variable)

Fixed Costs (fixed overhead and fixed SG&A costs)

- Does not change as the cost driver changes (assuming within the relevant range); outside of the relevant range or over time, may change
- Varies per unit; total remains constant

Variable Costs (direct materials, direct labor, variable overhead, and variable SG&A costs)

- Changes proportionally with the cost driver
- Constant per unit; total varies

Mixed Costs

- ~~Semi-variable costs (contain both fixed and variable components)~~

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D.1. Measurement Concepts: Part 1
D.3. Overhead Costs: Part 1

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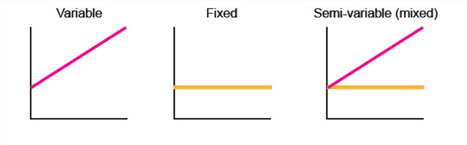
Illustration 8 Cost Behaviors



	Variable	Fixed	Semi-variable (mixed)
Sales	✓		
Less: returns and allowances	✓		
Cost of sales			
Direct material	✓		
Direct labor	✓		
Indirect labor			✓
Fringe benefits (15% of labor)	✓		✓
Royalties (1% of product sales)	✓		
Maintenance and repairs of building			✓
Factory production supplies	✓		
Depreciation: straight-line		✓	
Electricity: used in the mfg. process	✓	✓	
Scrap and spoilage (normal)	✓		
Selling, general, and administrative expense			
Sales commissions	✓		
Officers' salaries		✓	
Fringe benefits (relate to labor)	✓	✓	
Delivery expenses	✓		
Advertising expenses (annual contract expenses)		✓	

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Time Frame and Cost Classifications

- The relevant range is the range for which cost driver assumptions are valid.
- When the cost driver activity is outside of the range, variable and fixed cost assumptions for that driver cannot be used to allocate costs.
- In the long run, most costs will be variable.

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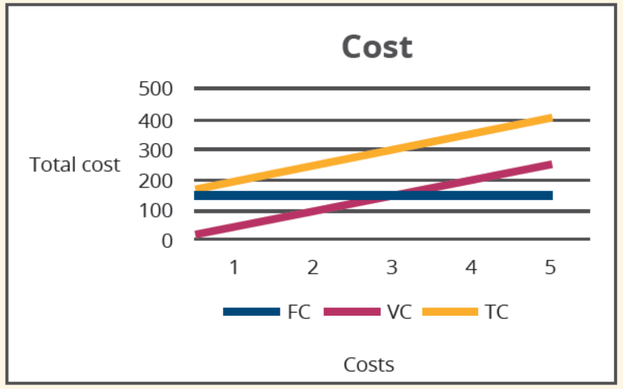
D.1. Measurement Concepts: Part 1
D.3. Overhead Costs: Part 1

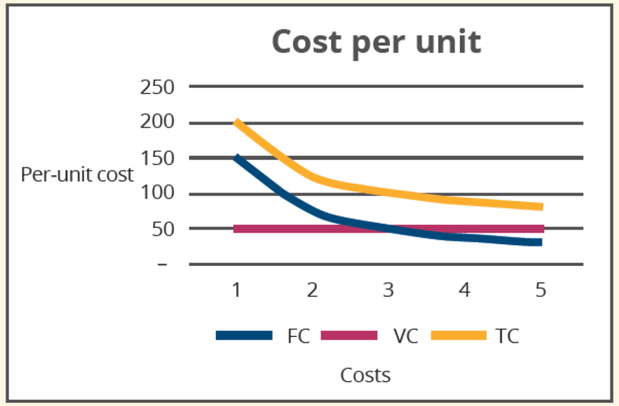
Illustration 9 Fixed and Variable Costs

In the illustration given below, the total fixed costs within a relevant range of 0 to 5 units is $150 and variable cost is $50 per unit.

Units	Fixed Cost (FC)	Variable Cost (VC)	Total Cost (TC)
1	$150	$ 50	$200
2	150	100	250
3	150	150	300
4	150	200	350
5	150	250	400



Within the relevant range, the total fixed cost remains constant at different level of outputs, but the variable cost increases with increase in output.



The fixed cost per unit decreases with increase in output, but the variable cost per unit remains constant. Within the relevant range, the total cost per unit decreases with increase in output.

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Analyzing Mixed Costs Using Regression Analysis

- Total costs are separated into their fixed and variable components for management to:
  - predict future cost behavior; and/or
  - facilitate operational decision making.
- Regression analysis is~~:~~
  - ~~one method used to separate mixed costs into their fixed and variable components;~~
  - ~~a statistical tool that explains cause and effect relationships and depicts this relationship in a linear function; and~~
  - ~~often used to depict the change in total production cost (effect) in response to a change in quantities (cause).~~

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Analyzing Mixed Costs

- Regression is used to estimate the relationship between the dependent variable and the independent variable or variables.
  - Simple linear regression involves one dependent variable (total costs) and one independent variable (volume or activity).

$$Y = a + bX$$

Or, **for cost separation purposes:**

Total cost = Fixed cost + (Variable cost per unit × Number of units)

Where:

Y = Total cost (the dependent variable)

a = Fixed cost (the Y-intercept)

b = The variable cost per unit (the slope of the line)

X = The number of units or any other cost driver (the independent variable)

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~~Analyzing Mixed Costs Using Regression Analysis~~

Illustration 10 Estimating Fixed Costs Using Regression Analysis

The accountant of a manufacturing company compiled the following information representing the level of production in units and the total cost for each of the previous twelve months:

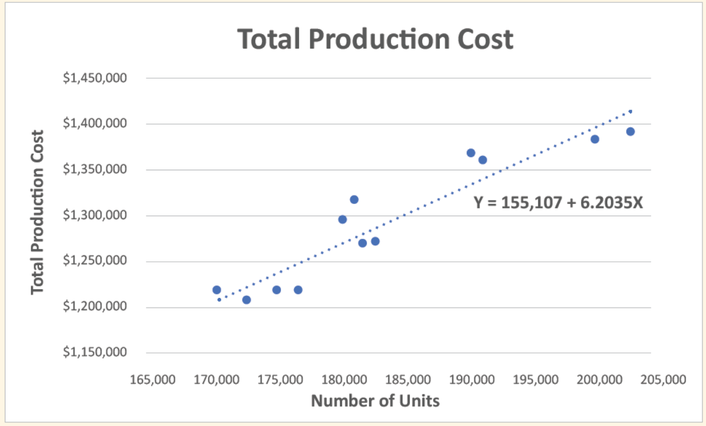
Month	Units of Production	Total Production cost
1	170,114	$ 1,218,298
2	180,916	$ 1,322,912
3	200,440	$ 1,385,580
4	190,095	$ 1,364,165
5	172,516	$ 1,208,112
6	190,977	$ 1,358,339
7	174,915	$ 1,217,905
8	180,012	$ 1,295,584
9	176,995	$ 1,218,465
10	202,519	$ 1,388,335
11	182,233	$ 1,276,232
12	181,415	$ 1,274,505

In this example, the total production cost is dependent on the number of units of production. The historical data shows the total production cost and the units of production. By inputting these values into statistical software, a regression equation is generated. In this equation, y= a +bX, a = the fixed costs (the y-intercept) and b = the variable cost per unit (slope of the line).

The output of the regression will give an estimate of fixed costs by solving for a value for a = fixed cost (the y-intercept). The output of the regression will also give an estimate of the variable costs by solving for a value for b = the variable cost per unit (the slope of the line).

Based on inputting the data above, the following results were generated by computer software:

	Coefficients
y-intercept	155,107
x-intercept	53.42



The regression equation was Y = $155,107 + 6.2035 X and the outputs are interpreted as follows:

- Fixed costs = the y-intercept or the constant coefficient, a, which is $155,107 per month.
- Variable cost per unit which is the variable coefficient or the x-variable, b, which is $6.20 per unit.

To apply this equation, a necessary assumption is that the level of activity falls within a relevant range of production. Within the relevant range, cost behaviors are assumed to remain the same. That is, total fixed costs will remain constant and variable costs per unit also will remain constant. Given these assumptions, the total cost can be estimated for any level of activity.

This function can be used to estimate total costs, as follows:

If the company estimates production for the coming month at a level of **220,000 units**, then the total estimated cost will be **$1,519,877** calculated as follows:

Total cost = $155,107 + ($6.2035 × 220,000 units)

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Variable and Fixed Overhead Allocation Using Cost Drivers

Cost Drivers and Costs Incurred

- Indirect costs are allocated to cost pools or objects using cost drivers (allocation bases) such as labor hours, machine hours, units produced, square footage, etc.
- Cost drivers should have a strong relationship to the incurrence of costs.
- Overhead may be allocated using traditional costing.
  - Step 1: Calculate the predetermined overhead rate (Budgeted overhead costs ÷ Estimated cost driver)

$$\text{Variable overhead application rate} = \frac{\text{Budgeted variable overhead}}{\text{Budgeted variable overhead cost driver}}$$

$$\text{Fixed overhead application rate} = \frac{\text{Budgeted fixed overhead}}{\text{Budgeted fixed overhead cost driver}}$$

  - Step 2: Apply overhead (Actual cost driver × Overhead rate)
- Activity-based costing (ABC) may also be used to allocate overhead.

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Variable Overhead Expenses: Allocation Bases

Examples:

- Direct labor hours or dollars
- Engineering or testing hours
- Machine hours
- Units produced

Example 3 **Variable Overhead Allocation Bases**

Facts: Henderson Corporation is analyzing its variable overhead expenses to determine the most accurate method of assigning variable overhead to each product. Henderson has identified the following variable overhead expenses:

1. Utilities
2. Repairs
3. Inspections

Required: Identify a potential cost driver for each of the above expenses.

Solution:

1. Utilities: Number of machine hours; the longer machines run the more utilities (for example, electricity for machines and heating or cooling the factory) may be used.
2. Repairs: Number of machine hours; more hours running the machines means more repair and maintenance time may be needed.
3. Inspections: Number of units produced; the more units produced, the more inspections on the units are needed.

Note: Total variable OH costs / Number of units produced = Variable OH cost per unit.

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Fixed Overhead Expenses: Allocation Bases

Examples:

- Budgeted usage (machine hours, labor hours)
- Budgeted level of service or operation (units of production expected)
- Physical capacity (total units that may be produced)
- Floor space (e.g., for building depreciation and property taxes)

Example 4 **Fixed Overhead Allocation Bases**

Facts: Henderson Corporation is assigning fixed overhead to each product. Henderson has identified the following fixed overhead expenses:

1. Factory rent
2. Factory depreciation
3. Factory supervisor's salary

Required: Identify a potential cost driver for each of the above expenses.

Solution:

1. Factory rent: square feet of factory
2. Depreciation: square feet of factory; allocating on a square-foot basis will assign more depreciation to products or divisions that use more factory space
3. Factory supervisor's salary: direct labor hours

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Overhead Allocation Methods

- Corporate-wide overhead (single overhead rate)
  - Applied to all departments using a volume-based cost driver

$$\text{Overhead application rate} = \frac{\text{Total budgeted overhead costs}}{\text{Total budgeted cost driver}}$$

- Departmental overhead (multiple overhead rates)
  - Unique cost drivers applied to each department

$$\text{Departmental overhead application rate} = \frac{\text{Budgeted department overhead costs}}{\text{Budgeted department cost driver}}$$

- Individual cost driver rate
  - Individual rates for different activities within a department

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Overhead Allocation Methods: Benefits and Limitations

- Corporate-wide overhead (single overhead rate)
  - Easy to use and allocation base is consistent across entire entity
  - Overhead costs assigned to activities may be over- or underestimated
- Departmental overhead (multiple overhead rate)
  - More accurate assignment of costs when there are multiple products produced in multiple departments
  - Requires identification and tracking of cost drivers by department
- Individual cost driver rate
  - Most accurate method of cost allocation
  - Complex and time-consuming

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D.1. Measurement Concepts: Part 1
D.3. Overhead Costs: Part 1

Example 5 **Overhead Rate Calculations**

Facts: Banner Corporation management estimates that the corporation's overhead for the year will total $500,000. Management is considering whether to use a corporate-wide overhead rate based on direct labor hours, to use departmental rates for the corporation's two production departments (Fabrication and Assembly), or to use individual rates based on individual cost drivers.

Departmental and individual cost drivers are as follows:

Departments	Cost Driver
Fabrication department	Machine hours
Assembly department	Direct labor hours

Individual	Cost Driver
Materials logistics	Material cost
Maintenance	Machine hours
Packaging	Number of shipments

Expected annual usage data follow:

	Fabrication	Assembly
Overhead	$200,000	$300,000
Direct labor hours	2,500	7,500
Machine hours	8,000	4,000
Materials costs	$480,000	$320,000
Number of shipments	N/A	10,000

The budgeted overhead cost for material logistics, maintenance, and packaging were $80,000, $300,000, and $120,000, respectively.

The following costs were incurred in March related to a new production run:

	Fabrication	Assembly
Direct labor hours	150	450
Machine hours	600	400
Materials costs	$50,000	$25,000
Number of shipments	N/A	750

Required: Calculate the overhead application rates assuming that Banner used (1) a corporate-wide rate, (2) departmental rates, and (3) individual cost driver rates (activities).

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Solution:

1. **Corporate-wide Rate:**

$$\frac{\$500{,}000 \text{ budgeted overhead costs}}{10{,}000 \text{ total budgeted direct labor hours}} = \$50/\text{direct labor hour}$$

2. **Departmental Rates:**

Fabrication: $\frac{\$200{,}000 \text{ overhead costs}}{8{,}000 \text{ machine hours}} = \$25/\text{machine hour}$

Assembly: $\frac{\$300{,}000 \text{ overhead costs}}{7{,}500 \text{ direct labor hours}} = \$40/\text{direct labor hour}$

3. **Individual (Activity) Rates:**

Material logistics: $\frac{\$80{,}000 \text{ overhead costs}}{\$800{,}000 \text{ material costs}} = \$0.10/\text{material dollar incurred}$ (10% of material costs incurred)

Maintenance: $\frac{\$300{,}000 \text{ overhead costs}}{12{,}000 \text{ machine hours}} = \$25/\text{machine hour}$

Packaging: $\frac{\$120{,}000 \text{ overhead costs}}{10{,}000 \text{ shipments}} = \$12/\text{shipment}$

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Cost Measurement Techniques

Actual Costing

- Uses actual costs incurred during production
- Used when cost measurement is required for a specific job
- Costs measured as follows:

Costs	Classification	Formula
Direct materials	Direct costs	Actual quantity × Actual price
Direct labor	Direct costs	Actual hours × Actual rate
Overhead	Indirect costs	Actual usage × Actual rate

- Most accurate measurement system
- Information is not timely because actual costs are not computed until after the conclusion of the production period
- Not useful for strategic planning or performance measurement
- Underutilized capacity situations cause significant fluctuations in per-unit costs

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Normal Costing

- Uses actual costs for direct materials and labor and normal costs for overhead
- Normal Costs: normally incurred during manufacturing

Costs	Classification	Formula
Direct materials	Direct costs	Actual quantity × Actual price
Direct labor	Direct costs	Actual hours × Actual rate
Overhead	Indirect costs (may use separate calculations for fixed and variable overhead)	Actual usage × Overhead application rate

- Overhead application computed by applying a predetermined overhead allocation rate

$$\text{Predetermined overhead rate} = \frac{\text{Total budgeted overhead}}{\text{Budgeted volume}}$$

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Normal Costing

- No fluctuations in overhead costs per unit
- No effect of over- or underutilized capacity on per-unit inventory costs
- Provides timely information regarding product costs
- Leads to variances because actual overhead incurred will likely differ from applied overhead

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Standard Costing

- Most common measurement system used
- Measures the costs the company expects to incur during production

Costs	Classification	Formula
Direct materials (for each item of material used)	Direct costs	Standard quantity allowed × Standard price
Direct labor (for each class of labor used)	Direct costs	Standard hours allowed × Standard rate
Overhead (may use separate calculations for fixed and variable overhead)	Indirect costs	Standard amount of allocation base for actual output × Standard rate

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Standard Costing

- Aids in cost control and reductions
- Provides information for performance evaluation, planning, and management by exception
- Time-consuming and may be subjective
- Differences between actual costs and standard costs will produce variances

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Example 6 Cost Computations

Facts: A company established the following standard costs per unit for a product:

- **Direct Materials:** 2 units of raw material @ $5 per unit
- **Direct Labor:** 3 direct labor hours @ $7 per hour
- **Variable Overhead:** $2 per hour applied on the basis of direct labor hours used to produce one item
- **Fixed Overhead:** $3 per hour applied on the basis of direct labor hours used to produce one item

During the period, the company manufactured 12,000 items with the following actual costs per item produced:

- **Direct Material:** 1.9 units of raw material @ $5.20 per unit
- **Direct Labor:** 3.5 direct labor hours @ $6.50 per hour
- **Variable Overhead Incurred:** $65,000
- **Fixed Overhead Incurred:** $100,000

Required: Compute the (1) actual, (2) normal, and (3) standard costs for the period.

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Solution:

1. Actual Costing:

Direct materials = Actual quantity × Actual price = 1.9 units actually used per each item produced × 12,000 items produced × $5.20 actual cost per unit = $118,560

Direct labor = Actual labor hours × Actual rate = 3.5 hours actually used for each item produced × 12,000 items produced × $6.50 actual cost per hour = $273,000

Variable overhead = $65,000

Fixed overhead = $100,000

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2. Normal Costing:

Direct materials = Actual quantity × Actual price = 1.9 units actually used per each item produced × 12,000 items produced × $5.20 actual cost per unit = $118,560

Direct labor = Actual labor hours × Actual rate = 3.5 hours actually used for each item produced × 12,000 items produced × $6.50 actual cost per hour = $273,000

Variable overhead = Actual labor hours × Standard rate = 3.5 hours actually used for each item produced × 12,000 items produced × $2 budgeted VOH per each DL hour = $84,000

Fixed overhead = Actual labor hours × Standard rate = 3.5 hours actually used for each item produced × 12,000 items produced × $3 budgeted FOH per each DL hour = $126,000

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3. Standard Costing:

Direct materials = Standard quantity allowed × Standard price = 2 units budgeted per each item produced × 12,000 units produced × $5 budgeted cost per unit = $120,000

Direct labor = Standard labor hours allowed × Standard rate = 3 hours budgeted for each item produced × 12,000 items produced × $7 budgeted cost per hour = $252,000

Variable overhead = Standard labor hours allowed × Standard rate = 3 hours for each item produced × 12,000 items produced × $2 budgeted VOH per each DL hour = $72,000

Fixed overhead = Standard labor hours allowed × Standard rate = 3 hours for each item produced × 12,000 items produced × $3 budgeted FOH per each DL hour = $108,000

Note that the overhead applied under standard costing and normal costing is different from the actual overhead incurred.

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